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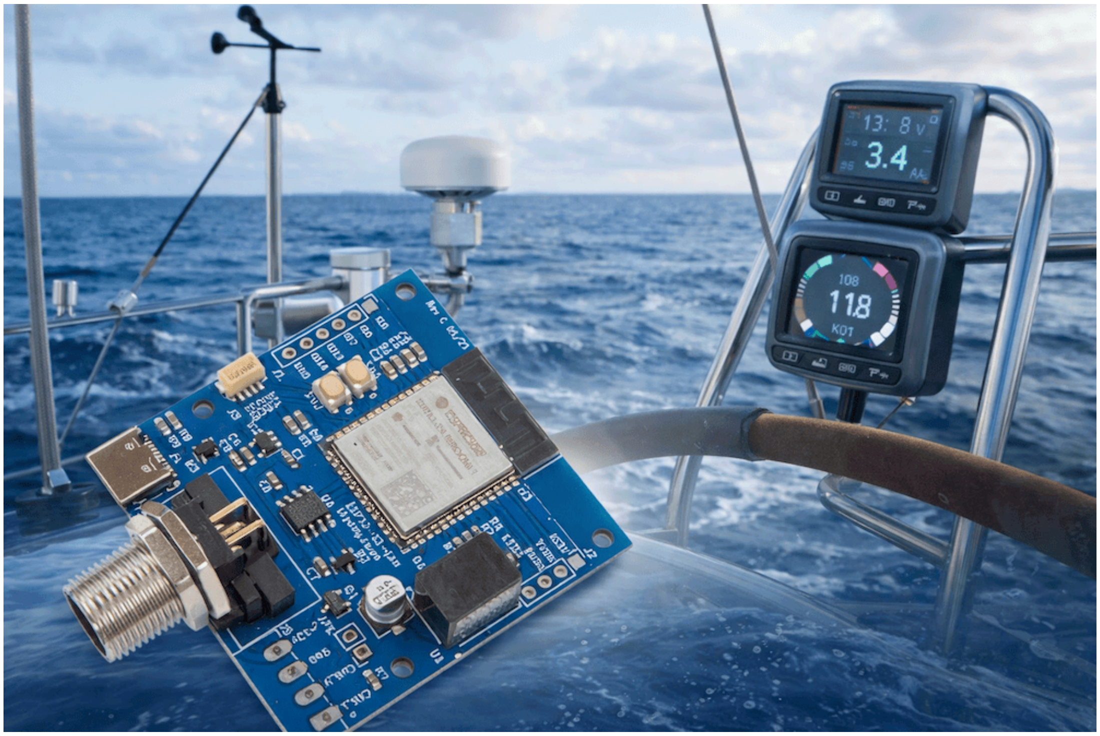
ESP32 NMEA 2000 Sensor Integration: Qwiic I2C Sensors for Marine Applications

Posted by Wilfried Voss on Jan 12th 2026



The ESP32S3 CAN-Bus Board with NMEA2000 Connector by Copperhill Technologies is a compact, high-performance development board based on the dual-core ESP32-S3-WROOM-1 microcontroller with integrated Wi-Fi and Bluetooth connectivity. Designed expressly for embedded and marine applications, it includes 8 MB of PSRAM and 8 MB of flash, a USB-C port for power and programming, RGB status LED, BOOT and RESET buttons, and a rugged supply range with reverse-polarity protection. Most importantly for marine electronics developers, it features a built-in CAN transceiver with an industry-standard NMEA 2000 Micro-C connector, allowing the board to plug directly into a vessel's NMEA 2000 backbone for seamless data exchange with GPS units, engine monitors, and other networked sensors.

In addition to its CAN/NMEA 2000 interface, **the board includes a Qwiic (I2C) connector for easy integration of external sensors** such as environmental and motion modules without soldering or

custom wiring. This makes it an ideal platform for prototyping or deploying custom sensor gateways, da
smart instrumentation on boats and other vehicles that use CAN-based networks. Whether you're captu
and power metrics, or attitude information from a motion sensor, this board serves as the central hub fo
distributing it over NMEA 2000 or forwarding it via Wi-Fi/Bluetooth to dashboards, apps, or cloud servic

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Please note that the devices listed below are suggestions only for use with the ESP32S3 CAN-Bus Boa
marine applications. Copperhill Technologies cannot provide technical support for third-party sensors o
support our own products, but we're not able to offer engineering or system design services.

We'll continue to publish blog posts and examples that highlight possible use cases and integrations. T
the extent of the engineering guidance we can provide for marine applications.

Qwiic-Compatible I2C Sensors for ESP32 and NMEA 2000 Applications

When designing an ESP32-based NMEA 2000 device, simplicity and reliability matter. The Qwiic ecosy
standardized 3.3 V I2C sensors with JST-SH connectors, onboard pull-ups, and no soldering required.
match for marine electronics prototypes, demo applications, and even production systems when senso

Below is a curated list of Qwiic-compatible sensors that work out of the box with an ESP32 and integrat
networks. Each section includes purchasing links and a suggested NMEA 2000 PGN mapping for docu

Environmental and Weather Sensors

BME280 – Temperature, Humidity, Barometric Pressure

The BME280 is a well-known, low-power environmental sensor and an ideal baseline for marine applica
climate monitoring and weather trend observation.

Purchase
SparkFun Qwiic BME280 Breakout

https://www.sparkfun.com/products/14348

Typical NMEA 2000 PGNs

- PGN 130310 – Environmental Parameters
  Temperature (air), humidity, barometric pressure

- PGN 130311 – Environmental Parameters (extended, optional)

Use cases

- Cabin temperature and humidity

- Weather trend display on MFDs

- Environmental logging

MS8607 – Temperature, Humidity, Pressure

The MS8607 combines a high-stability pressure sensor with humidity and temperature sensing. It is oft
accuracy and long-term stability are important.

Purchase
SparkFun Qwiic MS8607
https://www.sparkfun.com/products/16298

Typical NMEA 2000 PGNs

- PGN 130310 – Environmental Parameters
  Pressure, temperature, humidity

Use cases

- Weather monitoring
- Barometric trend alarms
- Environmental data logging

Power and Battery Monitoring

INA219 – Voltage, Current, Power
The INA219 is a simple current and voltage monitor suitable for basic battery or load measurements.

Purchase
SparkFun Qwiic INA219
https://www.sparkfun.com/products/15598

Typical NMEA 2000 PGNs

- PGN 127508 – Battery Status
  Battery voltage, current, temperature (if available)

Use cases

- House battery monitoring
- Load current measurement
- Entry-level power dashboards

INA226 – High-Accuracy Voltage and Current

The INA226 offers higher resolution and accuracy than the INA219 and is better suited for serious mari

Purchase
SparkFun Qwiic INA226
https://www.sparkfun.com/products/17704

Typical NMEA 2000 PGNs

- PGN 127508 – Battery Status
- PGN 127506 – DC Detailed Status (if implemented)

Use cases

- Battery charge/discharge tracking
- Solar or DC subsystem monitoring
- Power efficiency analysis

INA260 – Voltage and Current with Integrated Shunt

The INA260 includes an internal precision shunt resistor, simplifying wiring and installation.

Purchase
SparkFun Qwiic INA260
https://www.sparkfun.com/products/18279

Typical NMEA 2000 PGNs

- PGN 127508 – Battery Status

Use cases

- Compact battery monitoring solutions
- Demo and bench testing setups

Analog Sensor Interface

ADS1115 – 16-bit Analog-to-Digital Converter
Many marine sensors are analog by nature: tank level senders, pressure transducers, and NTC temper
bridges those sensors into the I2C/Qwiic world cleanly.

Purchase
SparkFun Qwiic ADS1115
https://www.sparkfun.com/products/15334

Typical NMEA 2000 PGNs

- PGN 127505 – Fluid Level
  Fuel, water, waste tanks
- PGN 130312 – Temperature
  External or fluid temperatures

Use cases

- Tank level monitoring
- External temperature probes

- Pressure transducer interfaces

Motion and Orientation Sensors

BNO055 – Absolute Orientation Sensor
The BNO055 includes onboard sensor fusion, making it one of the easiest ways to obtain pitch, roll, an
complex math.

Purchase
SparkFun Qwiic BNO055
https://www.sparkfun.com/products/14662

Typical NMEA 2000 PGNs

- PGN 127257 – Attitude
  Pitch and roll

- PGN 127250 – Vessel Heading (with caution)

Use cases

- Heel and trim monitoring

- Motion logging at anchor

- Stabilization or alert systems

ICM-20948 – 9-DoF IMU

A modern IMU offering accelerometer, gyroscope, and magnetometer data. Sensor fusion is handled in

Purchase
SparkFun Qwiic ICM-20948
https://www.sparkfun.com/products/15335

Typical NMEA 2000 PGNs

- PGN 127257 – Attitude

- PGN 127250 – Vessel Heading (if magnetometer used)

Use cases

- Advanced motion analysis

- Sailing performance metrics

- Custom attitude solutions

Light and Visibility

VEML7700 – Ambient Light Sensor
The VEML7700 measures ambient light levels and is useful for display dimming and day/night awarene

Purchase
SparkFun Qwiic VEML7700
https://www.sparkfun.com/products/14350

Typical NMEA 2000 PGNs

- PGN 130310 – Environmental Parameters
  Ambient light level (manufacturer-specific field if needed)

Use cases

- Automatic display dimming
- Daylight logging
- Night-mode switching

Digital Inputs and Expansion

MCP23017 – 16-bit GPIO Expander
While not a sensor itself, the MCP23017 is extremely useful in marine systems for reading multiple digi

Purchase
SparkFun Qwiic MCP23017
https://www.sparkfun.com/products/17022

Typical NMEA 2000 PGNs

- PGN 127501 – Binary Status Report
- PGN 126983 – Alert Response
- PGN 126985 – Alert Text

Use cases

- Bilge switch monitoring
- Hatch and door sensors
- Alarm and status inputs

Closing Notes

All sensors listed above operate at 3.3 V, support I2C natively, and connect directly via Qwiic without ac
suited for short cable runs and enclosure-mounted installations, which aligns well with typical marine el

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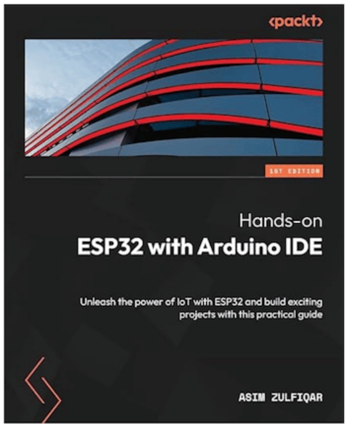
Hands-on ESP32 with Arduino IDE: Unleash the power of IoT with E
projects with this practical guide

The **ESP32** is a powerful and versatile microcontroller, ideal for those venturing into the world of **IoT (Ir**
wealth of capabilities—including Wi-Fi and Bluetooth connectivity, camera support, and sensor inter



integration with external components can be daunting for new **Integrated Development Environment (IDE)** simplifies pro access to ESP32's rich feature set, making it easier for beginner

This book is designed to guide you through the **fundamenta processing**, and real-world **IoT applications using the ESP** working with **ESP32 and Arduino IDE 2.0**, providing step-by-s development environment.

You'll then dive into **hands-on projects**, learning how to inte **ESP32-compatible camera and display modules**. These prac for understanding more advanced topics, such as **IoT network** WebSocket) and their roles in building connected devices.

As you progress, you'll apply your skills to build projects ranging from **smart devices** to **data loggers** a these engaging applications, you'll develop a solid understanding of how to design, prototype, and depl

By the end of this book, you'll be equipped to:

- Confidently develop and troubleshoot ESP32-based projects
- Choose appropriate IoT communication protocols for your applications
- Build and deploy functional IoT systems with real-world relevance

Whether you're a student, hobbyist, or aspiring engineer, this book provides a practical, project-base and its role in the ever-expanding world of IoT. More information...

ESP32 CAN BUS BOARD   ESP32 MARINE ELECTRONICS   ESP32 NMEA 2000   ESP32

ESP32 SENSOR GATEWAY   MARINE BATTERY MONITORING ESP32   MARINE CAN BUS

MARINE I2C SENSORS   MARINE MOTION SENSORS NMEA 2000   NMEA 2000 ENVIRONM

NMEA2000 PGN MAPPING   NMEA2000 POWER MONITORING   NMEA2000 SENSOR INTE

QWIIC I2C SENSORS   QWIIC SENSORS MARINE